

Lemma 4: let h be a level graph.

$\Rightarrow \underline{S(h)}$ is satisfiable

$(\Rightarrow) \underline{S(h^+)} \text{ is satisfiable}$

2SAT
on proper
 h

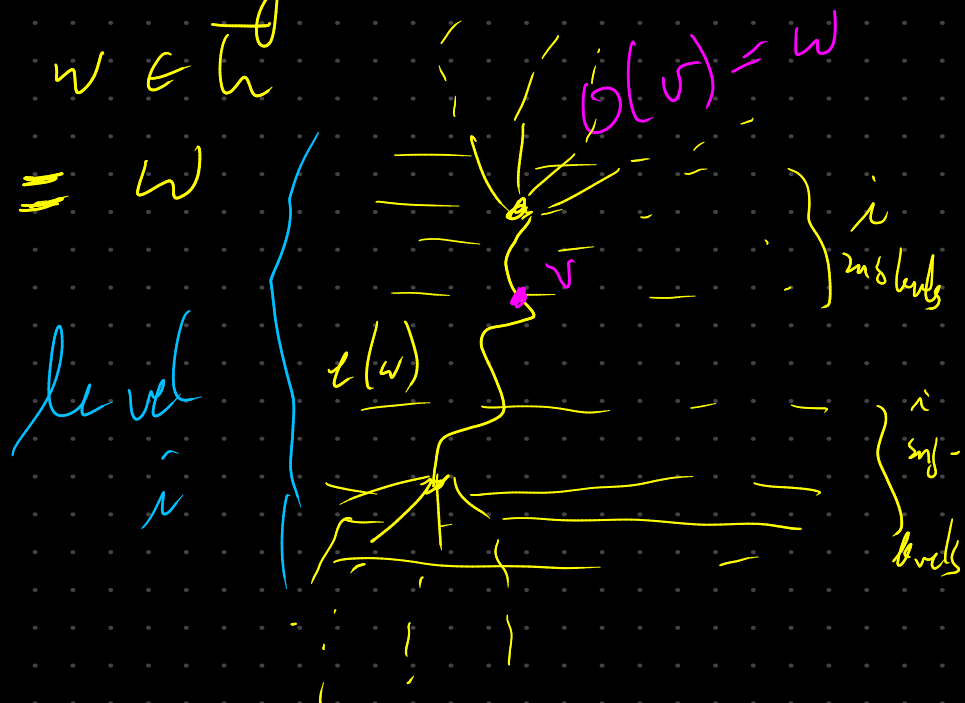
2SAT on
proper lit
induced from h^*

$\Rightarrow$ Assume there is a satisfying t.a. φ for $S(h)$
def: $O: V(h^+) \rightarrow V(h)$ (originate)

goal: describe φ^+ in relation to φ .

$\forall v \in V(h^+)$ and part of a stretch edge
 $e(w)$ of h^* for some
 $w \in h$

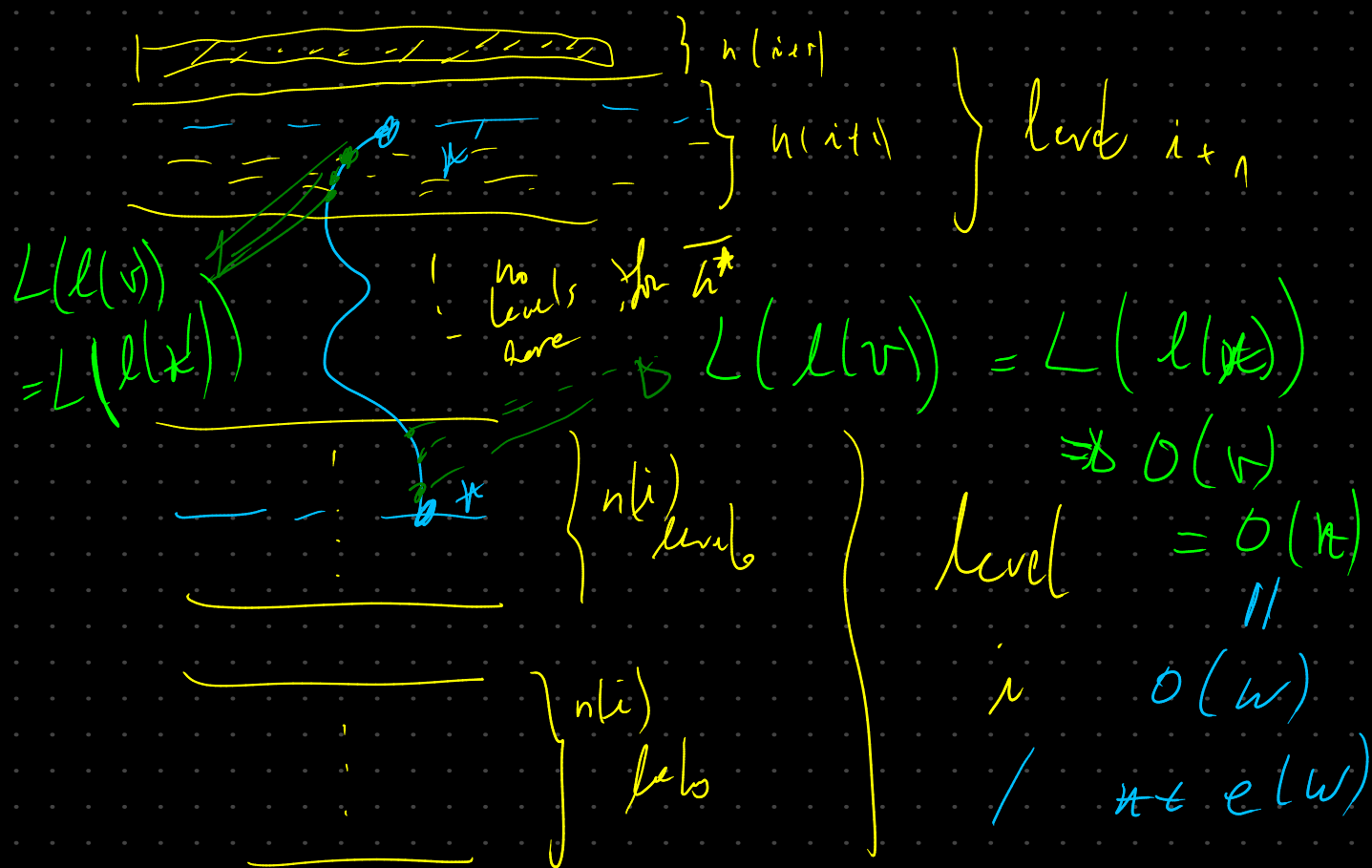
$\Rightarrow O(v) = w$



each other vertex $v \in V(\bar{h}^+)$ is a subdivision vertex of an edge (x, x') of h^* that is not a stretch edge.

map v to vertex $w \in \{x, x'\}$ with $L(l(v)) = L(l(w))$

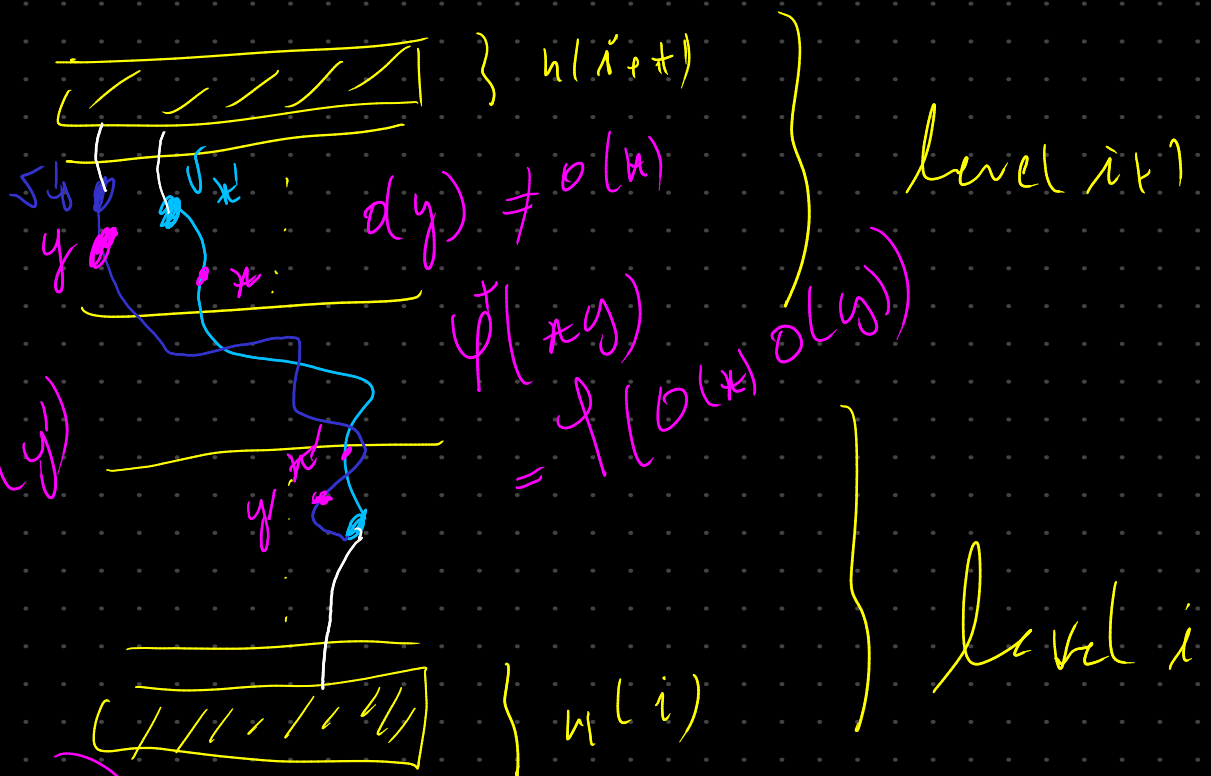
L maps each level (i, j) of h^* or h^+ to a level i of h



h distinct $x, y \in V(\bar{h}^+)$ with turns into (beside stretch edge) $\rightarrow$ one vertex

$$\left\{ \begin{array}{l} o(x) \neq o(y) \Rightarrow \\ \varphi^+(xy) = \varphi(o(x) o(y)) \end{array} \right.$$

$\left\{ \begin{array}{l} o(x) = o(y) \end{array} \right.$ they are both subdivision vertices of non-stretch edges e_x, e_y in T^* .



$$\begin{aligned} & \rightarrow o(x) = o(y) \\ & \varphi^+(x'y') \\ & = \varphi(o(x') o(y')) \end{aligned}$$

we define φ^+
consistency

$$\varphi^+ = f(\varphi) \text{ \& } \varphi \text{ consistent}$$

planarity

let $(u, v), (w, x) \in E(\bar{H}^+)$
with $l(u) = l(w)$

2e independent edges

if $l(l(u)) \neq l(l(v))$

$$\varphi^+(u, v) = \varphi^+(w, x)$$

level $i+1$

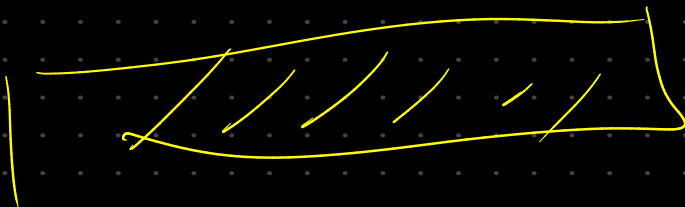
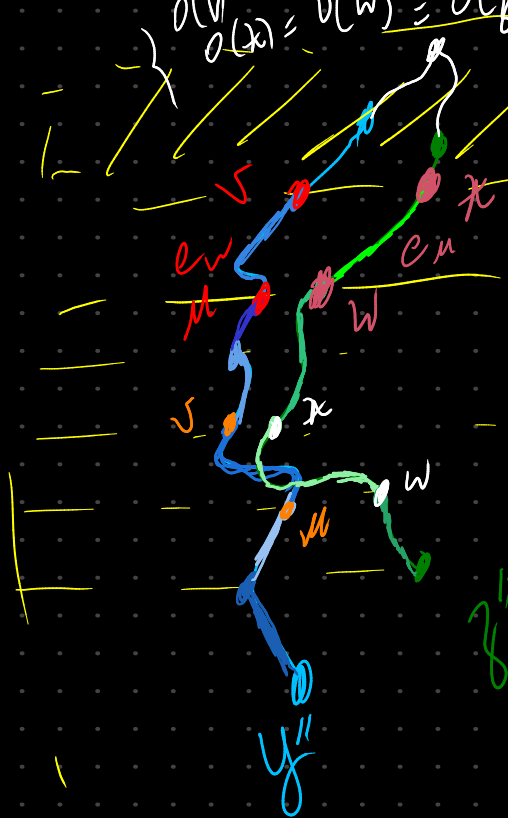
$$\varphi^+(u, v) = \varphi^+(w, x)$$

$$l(u) = l(w)$$

$n(i)$

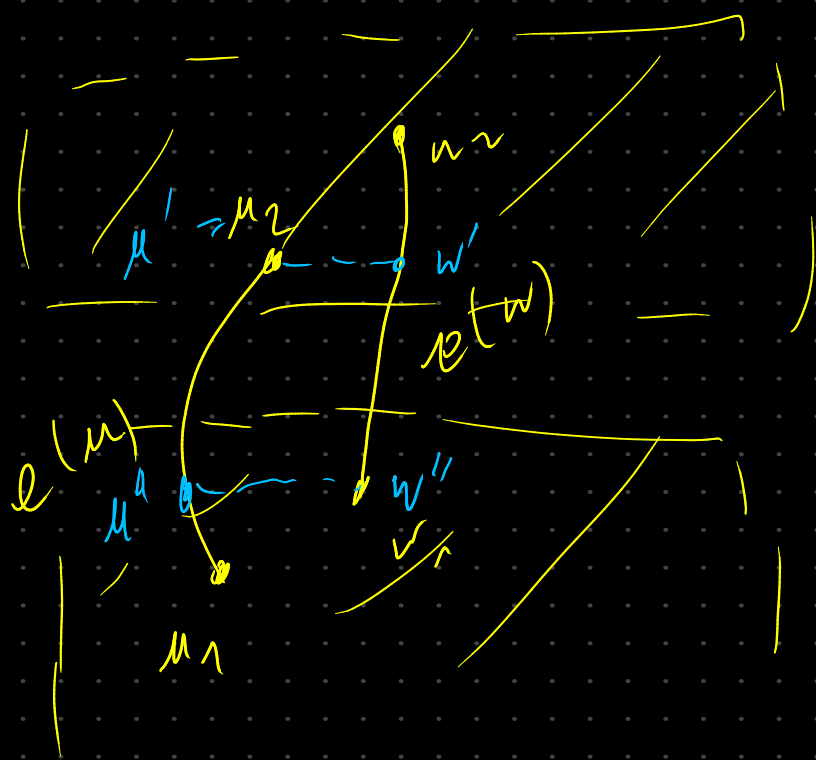
level i

$n(i)$



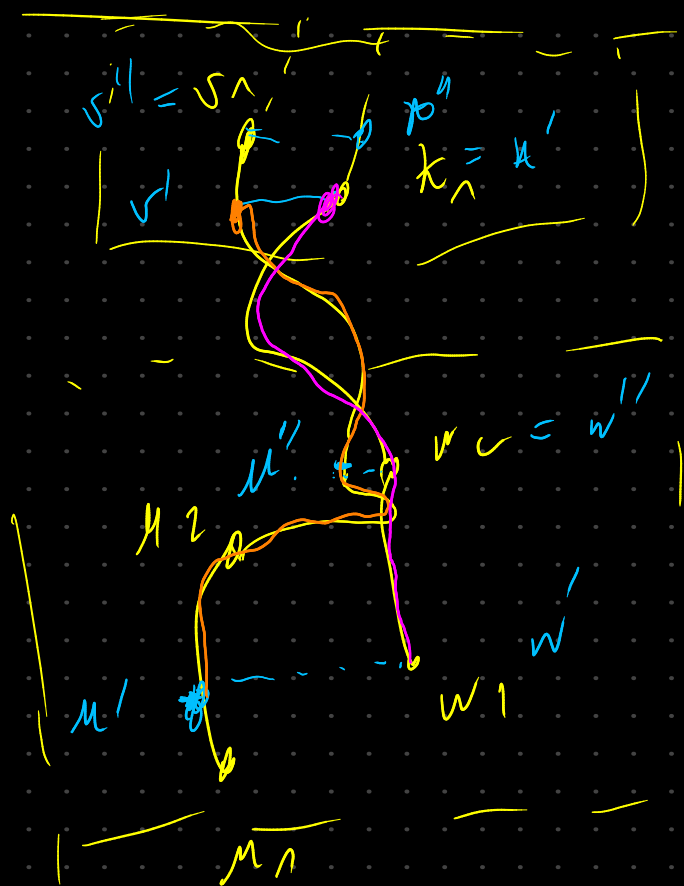
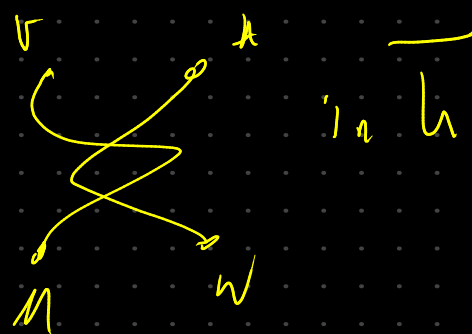
$\Leftarrow$) Assume $\exists$ satisfying t.n. φ^+ of $S(\overline{U}^+)$.

$u, w \in V(\overline{U})$ with $\ell(u) = \ell(w)$.



$$\varphi(uw) = \varphi^+(u'w')$$

$$\varphi^+(u'w') = \varphi^+(u''w'')$$



$$\varphi^+(u'w') = \varphi^+(v'w')$$

$\Downarrow$ (due to pre-order of φ)

$$\varphi^+(u'w') = \varphi(\overline{u'} \overline{w'})$$

$$\varphi^+(v'w') = \varphi(\overline{v'} \overline{w'})$$

$$\Downarrow$$

$$\boxed{\varphi(uw) = \varphi(vw)}$$

lemma 5: h level graph & s.t. a Γ^+ of $S(u^+)$
 $\Rightarrow \exists$ H-T drawing Γ^* of h^*

end of Extended PnPr

* IDEAS:

let h be a level graph

and $S(\bar{u})$ is satisfiable.

$\hookrightarrow$ create $S(\bar{u}^+)$

Level-transitivity
transitivity
 $\hookrightarrow$ Even crossing

$\hookrightarrow$ create H-T level drawing

(issue could be it's not cross)

$\hookrightarrow$ to weak H-T drawing

$\hookrightarrow$ x-monotone drawing (planar)

$$S(h) \rightarrow S(\bar{h}) \rightarrow S(\bar{h}^+)$$

H-T $\bar{h}^*$ of h^*

δ -monotone
drawing
of h

δ -monotone
of h^*

from h to h^* $\forall v \in V(\bar{h}) \exists! v_1, v_2 \in V(\bar{h}^*)$ s.t.
 $(v, v_1) = e(v)$

$$|V(\bar{h}^*)| = 2|V(\bar{h})|$$

we add sub-vertices to keep $\bar{h}^*$ proper $\rightarrow \bar{h}^+$

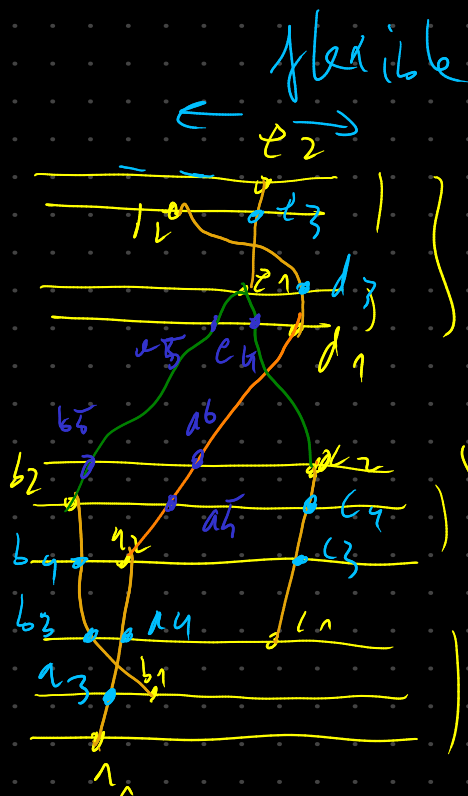
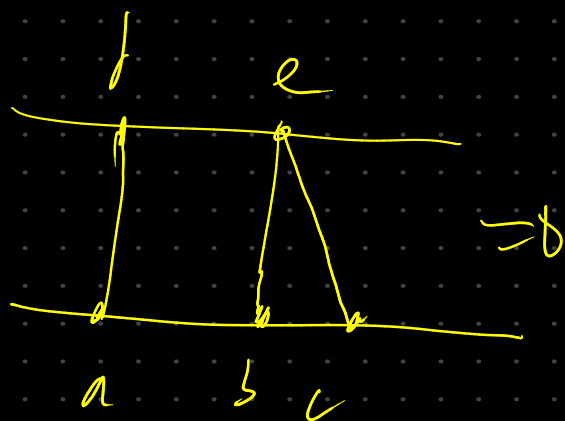
how to draw a H-T of $\bar{h}^+$?

$\forall$ level of h^+ change position based on ϕ^+

here a MAP when ever we make a new vertex
in $V(L^{+})$
and

we go level by level

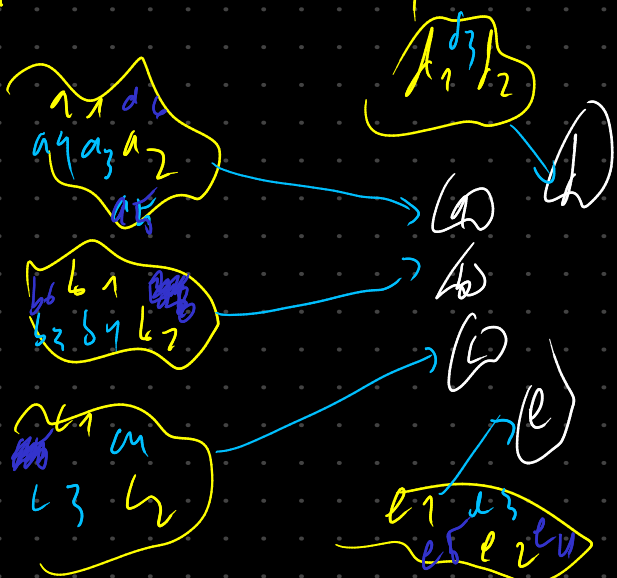
$$\varphi^+(xy) = \begin{cases} \varphi(o(x)o(y)) & \text{if } o(x) \neq o(y) \\ \varphi(o(v_x)o(v_y)) = o(y) & \text{if } o(x) = o(y) \end{cases}$$



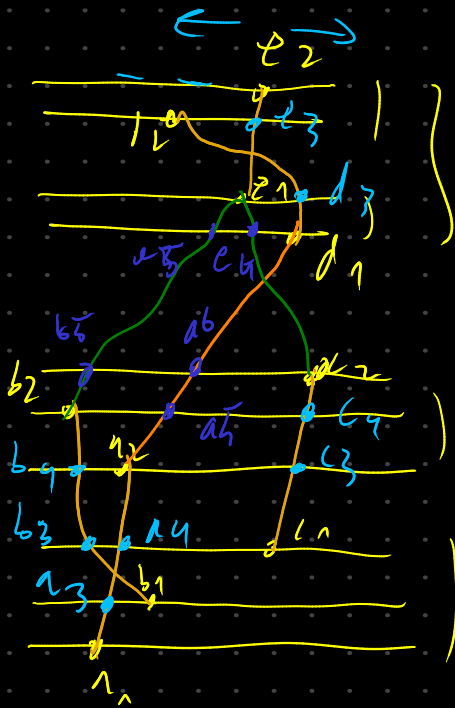
$$L(o(a_1)) + L(o(d_1))$$

$$(a_1, d_1)$$

$$\varphi \circ \varphi^+ \rightarrow \varphi$$



$$\varphi^+(xy) = \begin{cases} \varphi(o(x)o(y)) & \text{if } o(x) \neq o(y) \\ \varphi(o(v_x), v_y) & \text{else} \end{cases}$$



$$\varphi^+(a_3 b_1) = \varphi(ab) = 1$$

$$\varphi^+(c_1 b_3) = \varphi(cb) = 0$$

$$\varphi^+(c_1 a_4) = \varphi(ca) = 0$$

$$\varphi^+(a_2 b_1) = \varphi(ab) = 1$$

$$\varphi^+(a_2 c_3) = \varphi(ac) = 1$$

$$\varphi^+(b_2 a_5) = 0$$

$$\varphi^+(b_2 c_1) = 1$$

$$\varphi^+(c_2 b_1) = 0$$

$$\varphi^+(c_2 a_6) = 0$$

$$\varphi(ab) = 1$$

$$\varphi(cb) = 1$$

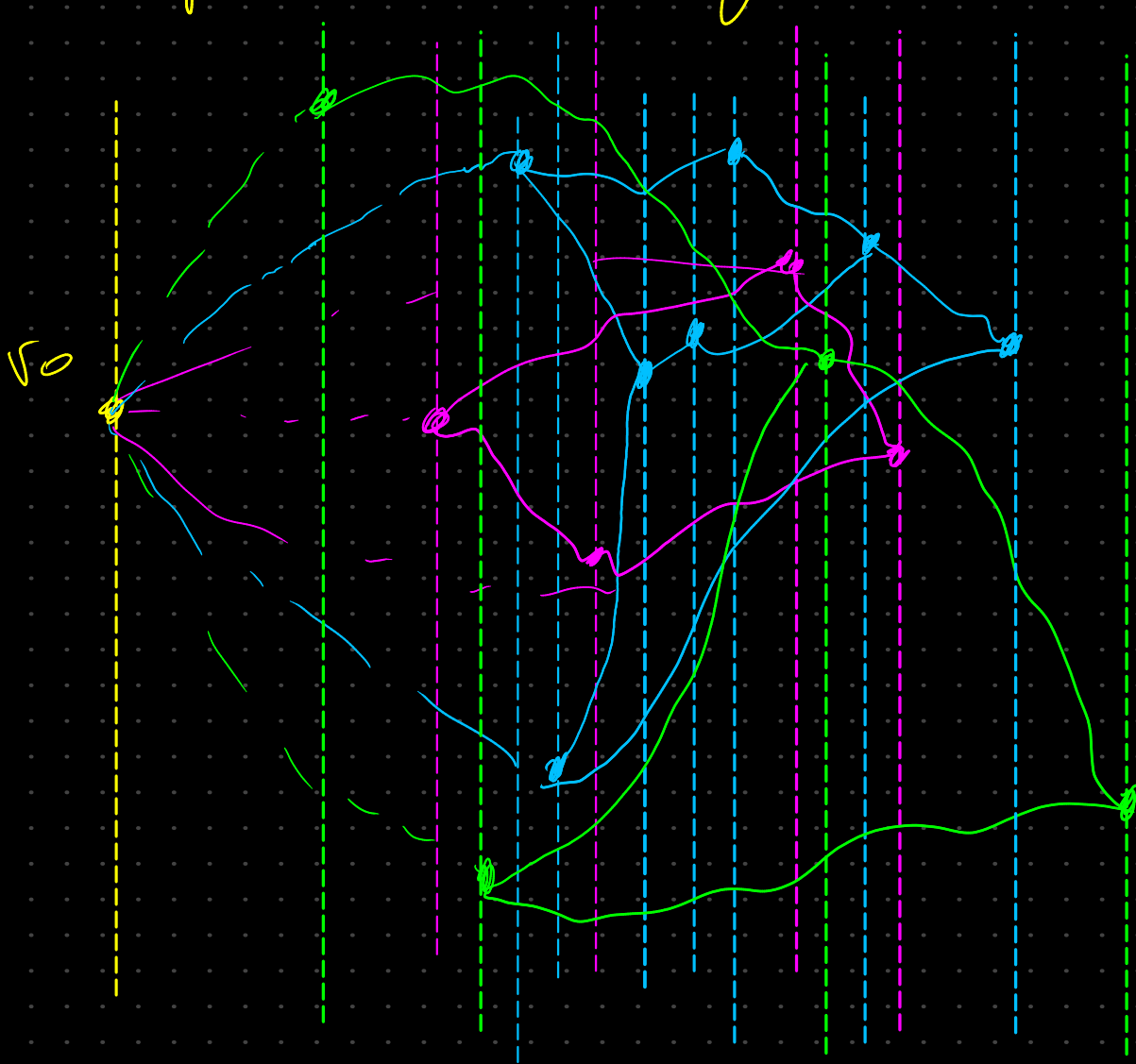
$$\varphi(ac) = 1$$

$$\varphi(ab) = 1$$

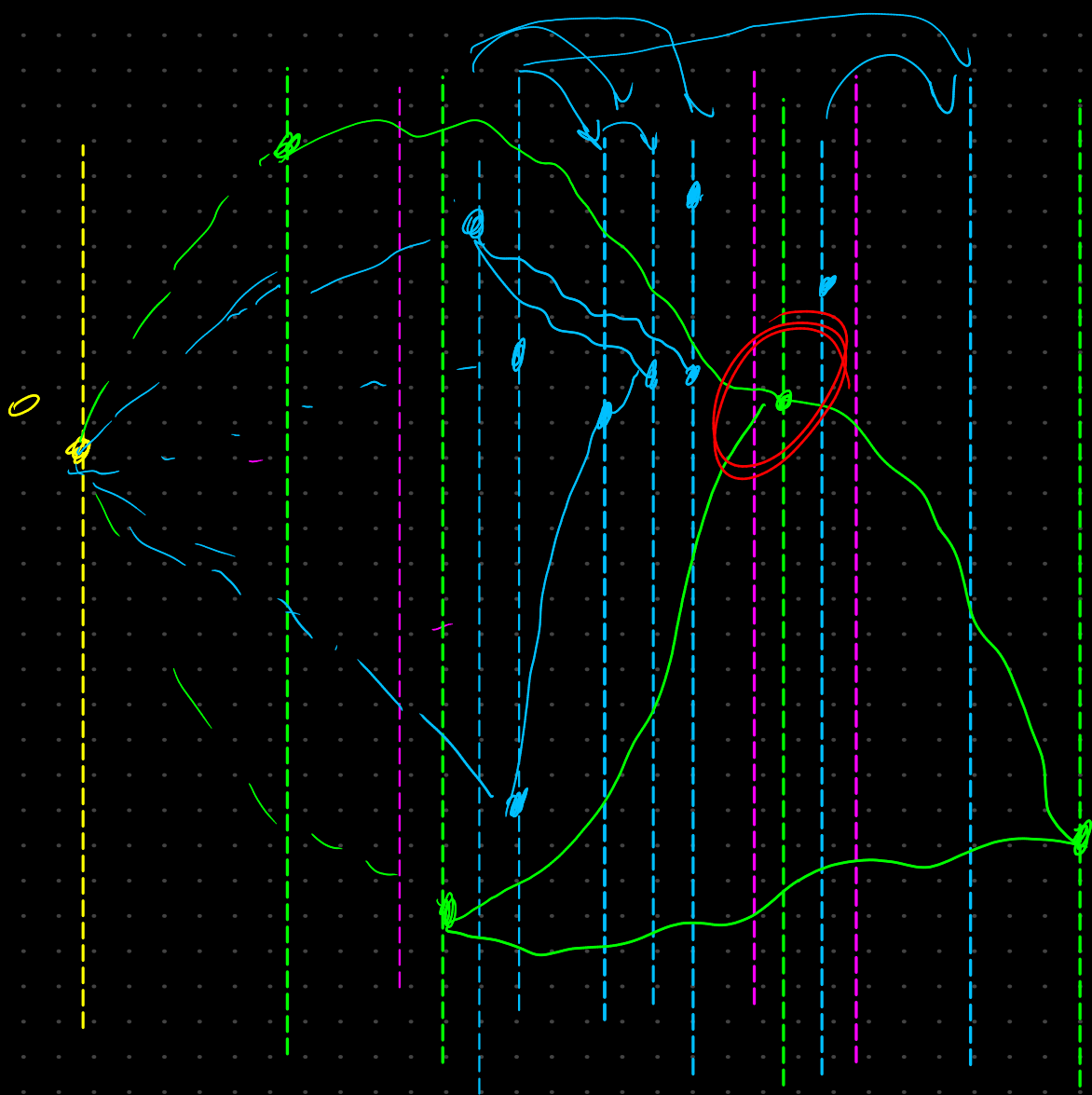


might have
issue
only with
adjacent edges.

trying to understand the impact of
conv. $h-A_1$ that gives multiple conv
(proof th 2.1) on my drawing steps



initial drawing



$v_1, \pm$ fixed the x -coord. even to
 the component are disjoint by fixing
 the x -axis. Basically he won't
 move here before adding x_1 .